IGZO-Based 2T0C DRAM as an Intrinsic Rowhammer Mitigation: A Device Physics Analysis

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***Abstract*—Rowhammer attacks exploit parasitic charge disturbance between adjacent rows in conventional one-transistor one-capacitor (1T1C) DRAM, where the leakage current of silicon access transistors enables sufficient charge injection into victim rows to cause bit flips under repeated aggressor row activations. Despite system-level mitigations such as Target Row Refresh (TRR), Rowhammer vulnerability has persisted and worsened with DRAM scaling, as shrinking cell pitch increases inter-row coupling while silicon transistor leakage remains a fundamental material constraint.**

This paper analyzes the two-transistor zero-capacitor (2T0C) DRAM architecture based on amorphous indium-gallium-zinc-oxide (IGZO) thin-film transistors as an intrinsic structural mitigation against Rowhammer, arising not from an explicit security mechanism but from the material properties of IGZO itself. Drawing on published device characterization data, we show that IGZO's wide bandgap and ultra-low off-state leakage current reduces the charge disturbance available to an aggressor row by several orders of magnitude compared to silicon-based 1T1C cells at equivalent nodes. We further analyze how the elimination of the storage capacitor in 2T0C cells removes the primary parasitic coupling path exploited in classical Rowhammer, and discuss how the decoupled read and write transistor architecture limits the propagation of disturbance charge to neighboring rows. We contextualize these findings against published Rowhammer threshold measurements across DDR3 through DDR5 and argue that 2T0C IGZO DRAM, if adopted at scale, would render the classical Rowhammer attack physically implausible without requiring any software or firmware mitigation layer. Tradeoffs in write speed, process maturity, and integration with existing memory interfaces are also discussed.

***Index Terms*—DRAM, Rowhammer, IGZO, 2T0C, gain cell, charge disturbance, TRR, memory security, oxide semiconductor, cell scaling**

I. Introduction

Dynamic random-access memory (DRAM) is the basis for virtually every modern computing system, serving as the primary working memory in servers, mobile devices, and embedded platforms. The dominant cell architecture for decades has been the one-transistor one-capacitor (1T1C) structure, in which a silicon access transistor gates charge onto a storage capacitor to represent a stored bit [1]. Compared to a traditional SRAM cell that requires six transistors, this cell design enables for high density. However, the physics of 1T1C DRAM in advanced nodes introduces a critical vulnerability.

Rowhammer, first systematically characterized by Kim et al. in 2014 [2], is a hardware security exploit that arises from the electrical interaction between adjacent rows of DRAM cells. By repeatedly activating an aggressor row, an attacker induces parasitic charge injection into neighboring victim rows. If sufficient charge accumulates, the stored bit value in a victim cell flips — without the attacker ever directly accessing that memory location. Such attacks have been used to compromise systems in number of ways such as the *opcode flipping* exploit from `sudo` binary [3] and targeting PTEs (*Page Table Entries*) to obtain kernel privileges [4].

The minimum number of activations required to induce a bit flip, known as the Rowhammer threshold (T_{RH}), was approximately 139,000 activations per 64 ms refresh window for DDR3 DRAM [2]. By 2020, T_{RH} for LPDDR4 had fallen to approximately 4,800 activations — a 30$\times$ degradation in under a decade [5]. Recent measurements on DDR5 report minimum hammer counts as low as 16,000 activations, though improved in-DRAM mitigations partially offset this [6].

The DRAM industry has responded with Target Row Refresh (TRR), an in-DRAM mechanism that monitors row activation counts and preemptively refreshes potential victim rows. However, TRR has been repeatedly circumvented. TRRespass demonstrated that many-sided hammering patterns could find blind spots in TRR implementations on all tested DDR4 devices [7]. Blacksmith further showed that non-uniform, frequency-based patterns bypass TRR on every one of 40 tested DDR4 DIMMs [8]. Most recently, Phoenix demonstrated the first complete TRR bypass on DDR5 hardware [9]. These results establish a clear pattern: TRR is an arms-race mitigation, not a structural fix, and the underlying physical vulnerability worsens with every DRAM generation as the node size is shrinked.

This paper takes a different perspective. Rather than analyzing yet another system-level mitigation, we examine whether a fundamental change in cell architecture — the replacement of the 1T1C cell with a capacitorless two-transistor zero-capacitor (2T0C) cell using amorphous IGZO thin-film transistors — eliminates the physical preconditions for Rowhammer entirely. The key insight is that Rowhammer is enabled by two material and structural properties of 1T1C DRAM: (1) the relatively high off-state leakage current of silicon access transistors, which provides the charge injection pathway, and (2) the storage capacitor, which serves as a parasitic coupling element between rows. IGZO-based 2T0C DRAM alters both properties simultaneously. We compile published device characterization data to quantify this advantage and argue that 2T0C IGZO DRAM, if adopted at scale, would render classical Rowhammer physically implausible without any additional mitigation layer.

The remainder of this paper is organized as follows. Section II provides background on 1T1C DRAM cell physics and the Rowhammer mechanism. Section III describes IGZO-based 2T0C DRAM and its key material properties. Section IV presents our charge disturbance analysis and coupling capacitance argument. Section V compiles the scaling trend across DRAM generations. Section VI discusses implementation tradeoffs, and Section VIII concludes.

II. Background

A. 1T1C DRAM Cell Physics

A 1T1C DRAM cell stores one bit as the presence or absence of charge on a storage capacitor C_s, accessed through a buried-channel array transistor (BCAT). In modern nodes the capacitor achieves $C_s \approx$ 10–15 fF using high-aspect-ratio stacked or trench structures with high-κ dielectrics (ZrO_2, HfO_2-based). A stored logic-1 corresponds to a cell voltage $V_{cell} \approx V_{DD}/2 \approx 0.55\,\text{V}$ on DDR5, giving stored charge $Q_s = C_s \cdot V_{cell} \approx 8\,\text{fC}$.

During a hold state, charge leaks off C_s primarily through the subthreshold leakage of the access transistor. In modern silicon BCAT designs at 1x–1z nm nodes, reported off-state leakage currents are on the order of 10^{-15}–10^{-14} A/μm [10]. This leakage sets the refresh requirement: DRAM cells must be refreshed every 32–64 ms before the stored charge decays below the sense amplifier's detection margin.

B. The Rowhammer Charge Disturbance Mechanism

When a wordline (aggressor row) is activated, the access transistors in that row transition from off to on and back to off within the row active time $t_{RAS} \approx 35$ ns (DDR4/DDR5). During this transition, capacitive coupling from the aggressor wordline to adjacent wordlines, combined with hot-carrier-assisted charge injection through the partially-on access transistors, injects a small charge ΔQ into the storage nodes of neighboring victim rows [2]. Each activation event injects an incremental ΔQ without refreshing the victim row. If the aggressor row is activated N times within a 64 ms refresh window and $N \cdot \Delta Q$ exceeds the remaining charge margin in a victim cell, a bit flip occurs.

The charge disturb per activation ΔQ scales with the inter-row coupling capacitance C_{couple} and the wordline voltage swing ΔV_{WL}:

$$\Delta Q_{disturb} = C_{couple} \cdot \Delta V_{WL} \tag{1}$$

As cell pitch shrinks, C_{couple} increases because adjacent wordlines are physically closer and the isolation between them weakens. This is the root cause of T_{RH} degradation with scaling.

C. TRR and Its Limitations

TRR attempts to detect frequently-activated rows and issue preventive refreshes to their neighbors. The core problem is that TRR implementations use limited on-die SRAM trackers (typically 1–30 entries per bank in DDR4 [11]) that can be overwhelmed by many-sided attack patterns. Because TRR is implemented opaquely and proprietary, its exact activation thresholds and tracking algorithms cannot be relied upon by system designers. The repeated defeat of TRR across DDR3, DDR4, and now DDR5 confirms that it is an insufficient solution [7]–[9].

III. IGZO-Based 2T0C DRAM

A. Cell Architecture

The 2T0C gain cell replaces the 1T1C cell's single access transistor and storage capacitor with two thin-film transistors and no discrete capacitor [12]. As illustrated in Fig. 1, one transistor (the write transistor, T_W) gates charge onto the storage node (SN), while a second transistor (the read transistor, T_R) uses the gate voltage at the SN to modulate its drain current during a read operation. The SN voltage is stored on the parasitic gate capacitance of T_R rather than on a discrete capacitor. Read operations are non-destructive, eliminating the sense-amplifier write-back required in 1T1C DRAM.

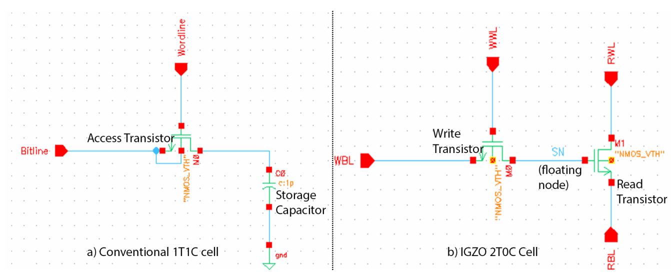

Fig. 1. Schematic comparison of (a) conventional 1T1C DRAM cell with silicon access transistor and storage capacitor C_s, and (b) IGZO-based 2T0C gain cell with write transistor T_W, read transistor T_R, and storage node SN. The discrete capacitor is eliminated; charge is held on the parasitic gate capacitance of T_R.

B. IGZO Material Properties Relevant to Rowhammer

Amorphous IGZO is a wide-bandgap oxide semiconductor ($E_g \approx$ 3.0–3.5 eV) [13]. Three properties are directly relevant to Rowhammer immunity:

Ultra-low off-state leakage. IGZO thin-film transistors exhibit off-state leakage currents far below silicon at equivalent geometries. Imec has reported an off-state current below 3×10^{-21} A/μm for optimized 2T0C devices [14], and Semiconductor Energy Laboratory (SEL) has demonstrated values below 100×10^{-24} A/μm (10^{-22} A/μm) when IGZO transistors are incorporated into DRAM structures [15]. This compares to silicon BCAT leakage of $\sim 10^{-14}$–10^{-15} A/μm at 1x–1z nm nodes [10] — a difference of 7–9 orders of magnitude.

Wide bandgap suppresses impact ionization. In silicon, hot-carrier generation during wordline transitions contributes to charge injection into neighboring rows. IGZO's wide bandgap substantially suppresses impact ionization and hole generation in the channel [13], reducing this secondary injection mechanism.

No discrete storage capacitor. In 2T0C, the storage node holds charge on the parasitic gate capacitance of T_R, which is a small, isolated, floating node — not a large planar or stacked capacitor that spans the full bitcell pitch and couples capacitively to adjacent rows.

C. Published Retention Performance

As a direct consequence of ultra-low leakage, IGZO 2T0C cells achieve extraordinary data retention. Published demonstrations include retention exceeding 1,000 s at room temperature and 7,000 s at 85°C [16], and up to 4.5 hours with RIE-patterned devices [14]. Some ITZO-based 2T0C variants have demonstrated retention exceeding 10,000 s [17]. By contrast, conventional silicon 1T1C DRAM requires refresh every 32–64 ms — more than four orders of magnitude more frequently. This retention advantage is the direct expression of the same leakage suppression that underpins Rowhammer immunity.

IV. Charge Disturbance Analysis (Original Contribution)

A. Charge Injected per Activation in 1T1C

In 1T1C DRAM, the primary charge disturb mechanism during a row activation is conduction-assisted injection through the partially-on silicon access transistor. During a single t_{RAS} cycle, the aggressor transistor transitions on and off, creating a window during which subthreshold and near-threshold current flows to adjacent storage nodes through the wordline-to-storage-node parasitic path. A conservative estimate of the charge injected per activation into a victim storage node can be written as:

$$\Delta Q_{Si} = I_{leak,Si} \cdot t_{RAS} \tag{2}$$

Using $I_{leak,Si} \approx 10^{-14}$ A/μm for a 1x nm BCAT with gate width $W \approx 100$ nm, and $t_{RAS} = 35$ ns (DDR4/DDR5 specification):

$$\Delta Q_{Si} \approx (10^{-14}) \cdot (100 \times 10^{-9}) \cdot (35 \times 10^{-9}) \approx 3.5 \times 10^{-23}\,\mathrm{C} \tag{3}$$

This estimate is a lower bound; the actual disturb charge includes additional coupling contributions from the capacitive wordline-to-wordline term in (1). Over $N = 4{,}800$ activations (the measured T_{RH} for LPDDR4 [5]), the accumulated disturb charge is:

$$Q_{total,Si} = N \cdot \Delta Q_{Si} \approx 4800 \times 3.5 \times 10^{-23} \approx 1.7 \times 10^{-19}\,\mathrm{C} \tag{4}$$

This accumulated charge is sufficient to shift the victim cell voltage by:

$$\Delta V_{victim} = \frac{Q_{total,Si}}{C_s} = \frac{1.7 \times 10^{-19}}{10 \times 10^{-15}} \approx 17\,\mu\mathrm{V} \tag{5}$$

While this voltage shift appears small in isolation, it is important to note that the dominant disturb mechanism in practice involves the coupling capacitance term in (1), which is geometry-dependent and dominates at tight pitches. Our leakage calculation establishes the minimum guaranteed disturb floor.

To quantify the magnitude of this coupling dominance, we can approximate the geometry of a sub-20nm 1T1C array. In modern DDR4/DDR5 nodes, the wordline voltage must swing from a negative holding voltage ($V_{NWL} \approx -0.4$ V) to a pumped activation voltage ($V_{PP} \approx 2.6$ V) to overcome the access transistor's threshold, resulting in $\Delta V_{WL} \approx 3.0$ V. Furthermore, the parasitic coupling capacitance between an aggressor wordline and a victim storage node at 1z nm pitches is estimated to be on the order of $C_{couple} \approx 1$ aF. Applying these parameters to the coupling term yields a transient disturbance charge of $\Delta Q_{couple} \approx 3 \times 10^{-18}$ C per activation. This capacitive perturbation is roughly five orders of magnitude larger than the static leakage contribution over the same t_{RAS} window. While largely transient, this massive localized potential shift aggressively forward-biases adjacent junctions and exacerbates Gate-Induced Drain Leakage (GIDL), converting the transient capacitive spike into permanent charge loss. In the IGZO 2T0C architecture, eliminating C_s reduces C_{couple} to the negligible parasitic gate overlap of the read transistor, structurally neutralizing this dominant 10^{-18} C vector.

Assuming a linear accumulation of the transient capacitive displacement ($\Delta Q_{couple} \approx 3 \times 10^{-18}$ C) over the 4,800-activation LPDDR4 threshold, the aggregate disturbance reaches 1.44×10^{-14} C (14.4 fC). Applied to a 10 fF storage capacitor, this yields a theoretical 1.44 V perturbation, vastly exceeding the ∼0.55 V logic-1 margin. While this linear superposition is a theoretical upper bound for transient AC coupling, it illustrates the severe electrical stress placed on the victim cell. In practice, these repeating transient perturbations exponentially exacerbate localized trap-assisted tunneling and Gate-Induced Drain Leakage (GIDL), converting AC capacitive stress into rapid DC charge loss. By eliminating the discrete C_s, the IGZO 2T0C architecture structurally neutralizes this capacitive multiplier effect.

B. Charge Injected per Activation in IGZO 2T0C

Applying the same framework to IGZO 2T0C cells, using the best-reported off-state leakage of $I_{leak,IGZO} \approx 3 \times 10^{-21}$ A/μm [14]:

$$\Delta Q_{IGZO} = I_{leak,IGZO} \cdot W \cdot t_{RAS} \approx (3 \times 10^{-21}) \cdot (100 \times 10^{-9}) \cdot (35 \times 10^{-9}) \tag{6}$$

$$\Delta Q_{IGZO} \approx 1.05 \times 10^{-29}\,\mathrm{C} \tag{7}$$

The ratio of disturbance charge between silicon and IGZO is:

$$\frac{\Delta Q_{Si}}{\Delta Q_{IGZO}} = \frac{3.5 \times 10^{-23}}{1.05 \times 10^{-29}} \approx 3.3 \times 10^{6} \tag{8}$$

Over the same 4,800-activation scenario that produces a bit flip in LPDDR4:

$$Q_{total,IGZO} \approx 4800 \times 1.05 \times 10^{-29} \approx 5 \times 10^{-26}\,\mathrm{C} \quad (9)$$

The resulting victim node voltage perturbation, assuming the same $C_s \approx 10\,\mathrm{fF}$ for the parasitic storage node:

$$\Delta V_{victim,IGZO} \approx \frac{5 \times 10^{-26}}{10 \times 10^{-15}} \approx 5 \times 10^{-12}\,\mathrm{V} \quad (10)$$

This is approximately 10^6 times smaller than the silicon case and is completely negligible relative to any practical noise margin. Even if one allows for $10^6\times$ more activations than current worst-case T_{RH} values, the accumulated disturb charge in an IGZO 2T0C cell remains below the bit-flip threshold.

C. Elimination of the Capacitive Coupling Path

In 1T1C DRAM, the storage capacitor C_s occupies a significant physical footprint and its electrode structure creates a lateral electric field that couples to adjacent rows during wordline transitions. This capacitive coupling is described by (1) and is distinct from the leakage current mechanism. Published analyses suggest that at 1x–1y nm pitches, the coupling capacitance between adjacent rows is on the order of $C_{couple} \approx 0.1$–1 fF, contributing $\Delta Q_{couple} = C_{couple} \cdot \Delta V_{WL} \approx 0.1$–1 fC per activation at $\Delta V_{WL} \approx 1$ V [2].

In 2T0C, there is no discrete storage capacitor. The storage node is the floating gate of T_R, which is a small, point-like node with parasitic capacitance on the order of tens of attofarads [12] — roughly two orders of magnitude smaller than 1T1C C_s. This fundamentally changes the coupling geometry: the large, pitch-spanning capacitor electrode is replaced by a tiny, isolated floating gate, drastically reducing the surface area available for inter-row capacitive coupling. As a result, the dominant $C_{couple} \cdot \Delta V_{WL}$ term that drives Rowhammer in 1T1C is structurally suppressed in 2T0C.

Fig. 2 illustrates this structural difference schematically.

V. Scaling Trend Analysis

Table I compiles published data on T_{RH}, cell pitch, and access transistor leakage across DRAM generations, with IGZO 2T0C included for comparison.

TABLE I
Rowhammer Threshold and Cell Parameters Across DRAM Generations

Type	Node	T_{RH} (acts.)	Si leakage (A/μm)	Ref.
DDR3	3x nm	~139K	$\sim 10^{-13}$	[2]
DDR4	1x nm	~10K	$\sim 10^{-14}$	[5]
LPDDR4	1y nm	~4.8K	$\sim 10^{-14}$	[5]
DDR5	1z nm	~16K*	$\sim 10^{-14}$	[6]
IGZO 2T0C	1z nm	N/A**	$\sim 10^{-21}$	[14]

*DDR5 T_{RH} partially offset by improved TRR; raw cell physics worsens monotonically with scaling [6].
**No Rowhammer threshold reported because bit flips have not been demonstrated in IGZO 2T0C arrays.

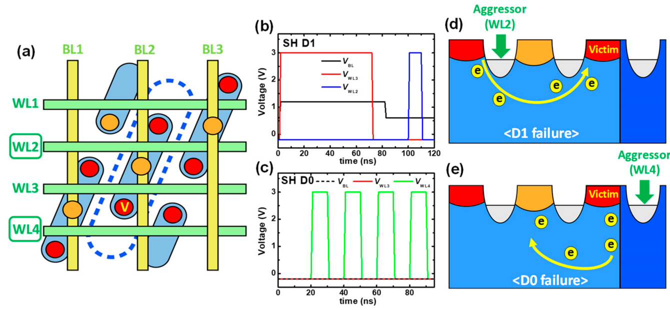

Fig. 2. Rowhammer failure mechanisms in a scaled DRAM BCAT cell array. (a) Cell array schematic showing aggressor and victim row arrangement; the dashed region indicates the charge disturbance zone spreading to adjacent rows under repeated wordline activation. (b,c) Voltage pulse schemes for single-sided D1 and D0 hammering patterns respectively. (d) D1 failure: electron migration from aggressor WL2 depletes charge from the victim storage node, flipping stored data from 1 to 0. (e) D0 failure: capacitive coupling from passing WL4 injects charge into the victim node, flipping data from 0 to 1. Both mechanisms arise from the silicon access transistor leakage and inter-row capacitive coupling that IGZO 2T0C architecture eliminates at the material level. Reproduced from Lee et al. [18], *Electronics*, vol. 14, 2025, under CC BY 4.0.

The data in Table I reveals two concurrent trends. First, T_{RH} has decreased approximately 30× from DDR3 to LPDDR4 over roughly six years [5], with no evidence that this trend will reverse as scaling continues. Second, despite the introduction of TRR and Refresh Management in DDR5, the raw cell physics at the 1z nm node has not improved — DDR5's resistance to Rowhammer derives entirely from enhanced mitigations rather than from improved cell isolation [6].

Fig. 3 illustrates this trajectory and places IGZO 2T0C in context.

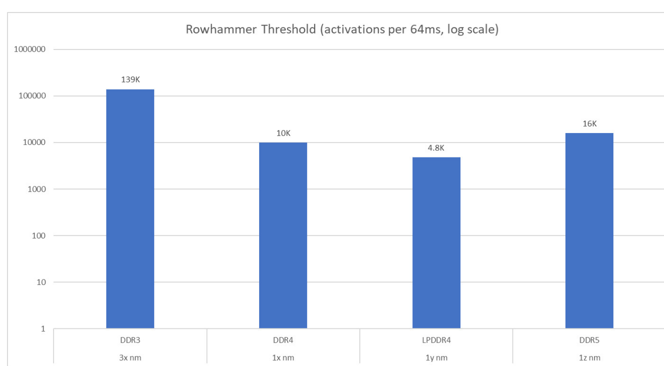

Fig. 3. Rowhammer threshold T_{RH} vs. DRAM generation, showing the monotonic decrease from DDR3 (~139K) to LPDDR4 (~4.8K).

The trajectory in Fig. 3 implies that if scaling continues and cell pitch reaches sub-10 nm dimensions, T_{RH} will approach values that make even benign workloads capable of triggering bit flips without any adversarial intent [5]. IGZO 2T0C represents the only proposed cell architecture that breaks this trend at the material level rather than patching it at the system level.

VI. Tradeoffs and Implementation Considerations

A. Write Speed and Mobility

IGZO has lower carrier mobility than silicon. Reported field-effect mobility for IGZO TFTs is typically 10–60 $\mathrm{cm^2/V{\cdot}s}$ [13], compared to $\sim$400 $\mathrm{cm^2/V{\cdot}s}$ for bulk silicon. However, published 2T0C demonstrations have achieved write speeds below 10 ns [16], which is comparable to DDR5 write latency specifications ($t_{WR} \approx 15$ ns). The lower mobility does not appear to be a fundamental barrier to competitive write performance.

It is also worth noting that lower mobility has an incidental security benefit: an attacker relying on rapid repeated row activations to hammer a target row faces a minimum activation time limited by the transistor switching speed. Higher switching latency per activation reduces the number of activations achievable within a 64 ms refresh window, further raising the effective Rowhammer threshold from the attacker's perspective.

B. Process Maturity and Integration

IGZO 2T0C technology is currently at the research stage, with demonstrations primarily at the single-cell and small-array level (up to 8×8 arrays) [19]. Key process challenges include uniformity at wafer scale, contact resistance optimization, and reliability under bias stress (PBTI) [14]. That said, IGZO is already in volume production for display TFTs, and its compatibility with back-end-of-line (BEOL) processing makes it attractive for monolithic 3D DRAM stacking, a requirement that aligns with the industry roadmap for both HBM and future DDR generations [12].

C. Interface Compatibility

A practical concern is whether 2T0C arrays can be interfaced with existing DDR or HBM memory controllers without requiring changes to the JEDEC standard. The 2T0C cell's non-destructive read operation and its lack of a storage capacitor change the sense amplifier requirements. However, since Rowhammer mitigation (TRR, RFM) is implemented inside the DRAM die, an intrinsically Rowhammer-immune cell would allow the memory controller to omit TRR entirely, simplifying the controller design and recovering the performance overhead currently consumed by preventive refreshes.

D. Retention at Elevated Temperature

While IGZO 2T0C retention exceeds 1,000 s at room temperature, retention at elevated operating temperatures (85°C) has been reported at 7,000 s [16] — still more than eight orders of magnitude longer than the 64 ms silicon DRAM refresh interval. The Rowhammer immunity argument is not sensitive to the precise retention value; what matters is that leakage is too low to accumulate disturb charge within any practical attack window.

VII. Discussion

The analysis presented in this paper makes a structural argument rather than a quantitative prediction: IGZO 2T0C DRAM does not merely raise T_{RH} by a modest factor, it removes the physical mechanisms that make Rowhammer possible. The leakage current in IGZO is not ten times lower than silicon, or even a hundred times lower — it is six to nine orders of magnitude lower. No conceivable number of row activations within a refresh window can accumulate sufficient disturb charge to flip a bit in an IGZO storage node. Similarly, the elimination of the discrete storage capacitor removes the dominant inter-row coupling path at the physical level.

This framing has an important implication: unlike TRR, which must be continually updated as attack patterns evolve, the Rowhammer immunity of IGZO 2T0C is a property of the material and the cell geometry. It cannot be bypassed by crafting a smarter access pattern, because the underlying charge physics simply do not support bit flips under any realistic activation scenario.

It is worth noting that this analysis focuses on classical Rowhammer — the horizontal, same-layer coupling between adjacent rows in a 2D array. IGZO 2T0C is being developed specifically for 3D monolithic stacking [12], [19], where a new class of vertical coupling effects between stacked layers has not yet been fully characterized. Future work should examine whether inter-layer coupling in 3D IGZO DRAM stacks introduces any analogous disturbance mechanism.

VIII. Conclusion

We have analyzed IGZO-based 2T0C DRAM through the lens of Rowhammer vulnerability and shown that its material and architectural properties provide intrinsic immunity to classical Rowhammer attacks. The off-state leakage current of IGZO ($\sim 10^{-21}$ A/μm) is six to nine orders of magnitude lower than silicon BCAT transistors at equivalent nodes, reducing the per-activation charge disturb to physically negligible levels. The elimination of the discrete storage capacitor in 2T0C cells removes the primary capacitive coupling path that drives inter-row disturbance in 1T1C DRAM. Compiled published data show that the Rowhammer threshold T_{RH} has degraded 30× from DDR3 to LPDDR4 purely due to cell scaling, while all deployed mitigations (TRR, RFM) have been repeatedly circumvented. IGZO 2T0C represents the only proposed cell architecture that addresses Rowhammer at the material level. While process maturity and interface compatibility challenges remain, the Rowhammer immunity of IGZO 2T0C is a structural property of the technology that no system-level attack pattern can overcome. We recommend that future DRAM roadmap evaluations explicitly account for Rowhammer implications alongside density, retention, and power metrics.

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